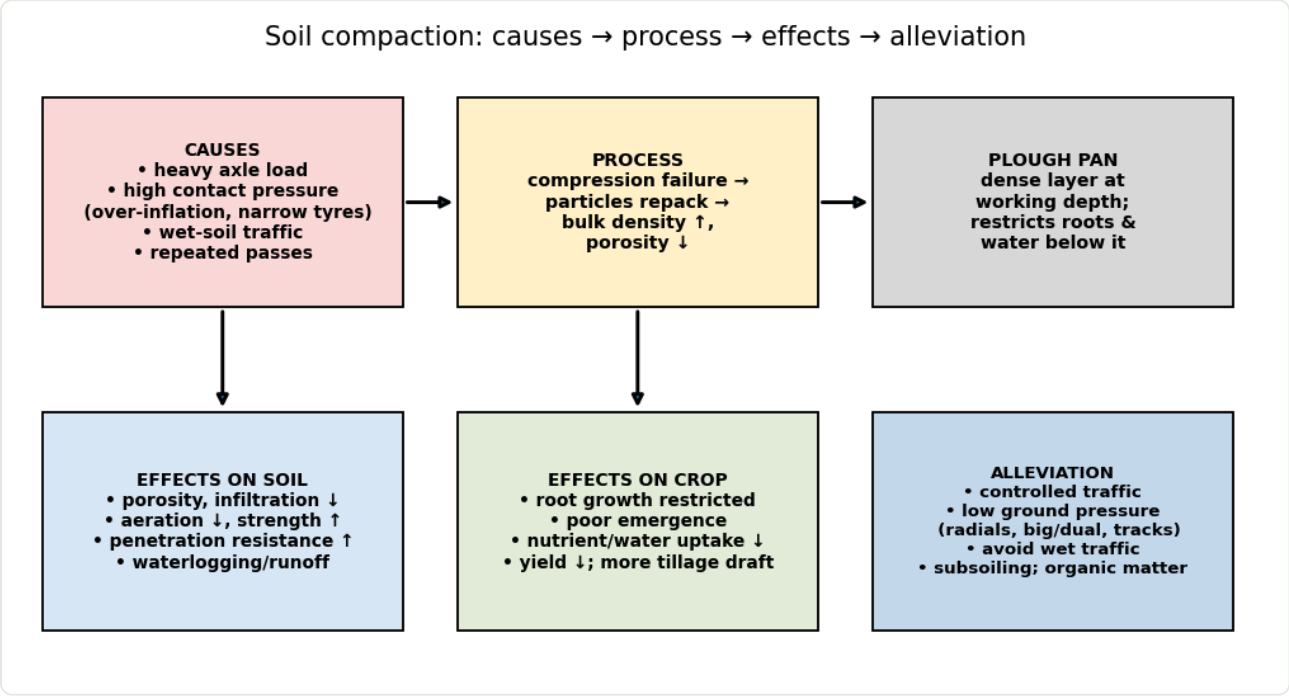
UNIT 6 · TOPIC 9(A)

Soil Compaction — What It Is and What Causes It

Syllabus topic: Soil compaction: causes, methods of alleviation, and effects on soil and crop responses

What soil compaction is

- Soil compaction is DENSIFICATION — a reduction in volume (of the air-filled pore space) under load: bulk density rises and porosity falls.
- In soil-dynamics terms (Topic 2) it is soil FAILURE BY COMPRESSION (volume change), unlike the shear failure a tillage tool causes. The same soil element may SHEAR under a narrow tine yet COMPACT under a wheel.
- It is largely reversible only by tillage/weathering, not spontaneously — a lasting problem.



Soil compaction: causes → process → effects → alleviation

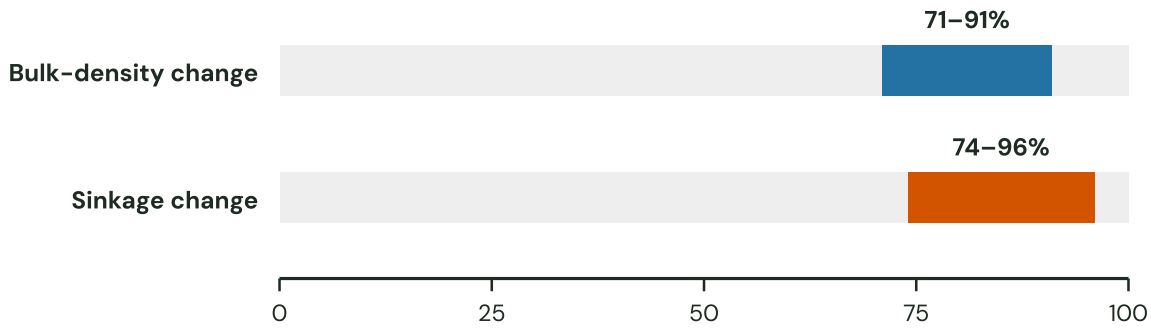
Causes

Mechanical — the running gear of tractors, trailers, combines and implements — is the dominant cause on farms. Key drivers:

Driver	Governs
High AXLE LOAD	How DEEP compaction reaches — subsoil compaction from heavy axles is persistent
High CONTACT (ground) PRESSURE — over-inflated or narrow tyres	How SEVERE the topsoil compaction is
SOIL MOISTURE at trafficking	Wet soil is far more compactible (Proctor curve)
NUMBER OF PASSES	Repeated wheeling in the same track

Contributory causes: natural (raindrop impact, wetting–drying, poor structure/low organic matter), tillage pans from repeated working at the same depth, and puddling in paddy (a deliberate compaction of the subsurface to hold water).

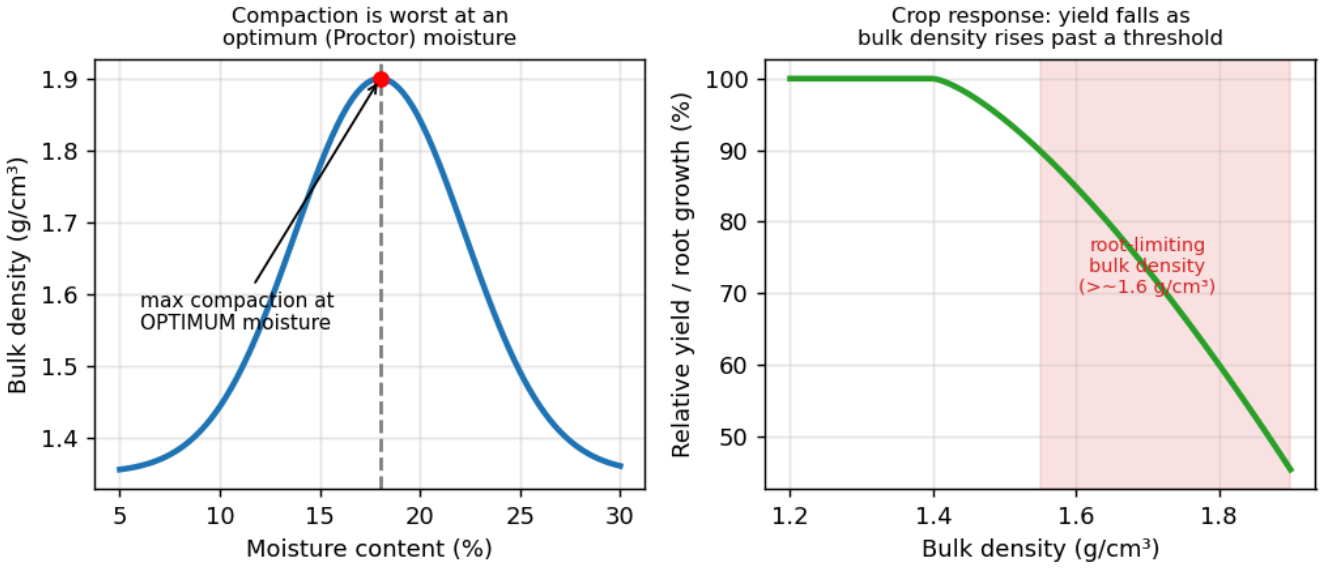
The first pass does most of the damage (Liljedahl Table 10-3, p. 257)



First-pass compaction — 13.6-38 bias six-ply R-1 tyre at 132 kPa (redrawn showing only the ranges stated in Table 10-3)

- After the FIRST pass, bulk-density change was **71–91%** and sinkage change **74–96%** of the TOTAL change measured after FOUR passes in the same rut.
- **Practical lesson:** the first wheeling compacts — keep traffic in defined lanes (controlled traffic) so most of the field is never wheeled.
- Compaction and traction are linked: a front wheel's compaction can give the rear wheel a better traction surface (p. 256).

The Proctor relation — moisture and compactability



Proctor curve (left) and crop response to bulk density (right) — illustrative curves

- For a given compactive effort, bulk density rises with moisture to a MAXIMUM at an OPTIMUM moisture content, then falls.
- **Why:** dry soil resists rearrangement (high strength); very wet soil has pore water carrying the load (it shears/smears instead); at the optimum, water lubricates particles into their densest packing.

- **Field consequence:** soil is MOST vulnerable near this optimum (roughly the field-capacity/plastic range) — exactly when farmers are tempted to work it.

> **Safe rule:** keep heavy traffic OFF the field when it is wet.

Practice MCQs

Q105. Most soil compaction from a wheel occurs during the:

(a) First pass
(b) Tenth pass
(c) Turning
(d) Braking

Q106. Soil compaction is best described as:

(a) Loss of soil water
(b) Densification — bulk density rises and porosity falls
(c) Erosion
(d) Salinity build-up

Q107. In soil-dynamics terms compaction is soil failure by:

(a) Shear
(b) Compression (volume change)
(c) Tension
(d) Melting

Q108. Under a wheel the soil typically fails by compression, whereas under a narrow tine it fails by:

(a) Compression
(b) Shear
(c) Evaporation
(d) No failure

Q109. The DEPTH to which compaction reaches is governed mainly by:

(a) Contact pressure
(b) Axle load
(c) Tyre colour
(d) Speed

Q110. The SEVERITY of surface compaction is governed mainly by:

(a) Axle load
(b) Contact (ground) pressure — inflation and tyre width
(c) Engine power
(d) Slip

Q111. Soil is MOST compactible when it is:

(a) Oven dry
(b) At/near an optimum (field-capacity) moisture
(c) Saturated
(d) Frozen

Q112. According to Liljedahl's Table 10-3, most of the compaction (71–96% of the 4-pass total) occurs on the:

(a) Fourth pass
(b) First pass
(c) Turning
(d) Braking

Q113. The practical lesson from the first-pass finding is to:

(a) Wheel everywhere
(b) Confine traffic to permanent lanes (controlled traffic)
(c) Inflate tyres more
(d) Add ballast

Q114. A dense sub-surface layer formed at the usual tillage depth is called a:

(a) Crust
(b) Plough (hard) pan
(c) Furrow slice
(d) Mulch

Q115. The curve of bulk density versus moisture at fixed effort is the:

(a) Mohr circle
(b) Proctor curve
(c) Willan line
(d) NDVI curve

Q116. On the Proctor curve, maximum bulk density occurs at:

(a) Zero moisture
(b) An optimum moisture content
(c) Saturation
(d) Wilting point

Q117. Very wet soil resists compaction to maximum density because:

(a) It is strong
(b) Pore water carries the load and the soil shears/smears
(c) It is frozen
(d) No particles present

Q136. Puddling in paddy fields is a deliberate:

(a) Loosening
(b) Compaction of the subsurface to reduce percolation
(c) Drying
(d) Levelling only

Q137. Compaction and traction are linked because:

(a) A firmed surface can improve a following wheel's traction
(b) They are unrelated
(c) Compaction removes traction
(d) Traction dries soil

Q138. Over-inflating a tractor tyre tends to:

(a) Reduce surface compaction
(b) Increase contact pressure and surface compaction
(c) Increase footprint
(d) Lower draft

Answers

Q105 — (a) First pass

Q106 — (b) Densification — bulk density rises and porosity falls

Q107 — (b) Compression (volume change) — *Topic 2 distinction — compression (wheel) vs shear (tine) failure of the SAME soil.*

Q108 — (b) Shear

Q109 — (b) Axle load — *Axle load sets the DEPTH of the compacted zone.*

Q110 — (b) Contact (ground) pressure — inflation and tyre width — *Contact pressure (inflation, tyre width) sets the SEVERITY near the surface — a standard exam pair.*

Q111 — (b) At/near an optimum (field-capacity) moisture

Q112 — (b) First pass — *Liljedahl Table 10-3, p. 257 — first pass = 71–91% (bulk density) and 74–96% (sinkage) of the 4-pass total.*

Q113 — (b) Confine traffic to permanent lanes (controlled traffic) — *Hence controlled traffic.*

Q114 — (b) Plough (hard) pan

Q115 — (b) Proctor curve — *Proctor relation — densest packing at optimum moisture.*

Q116 — (b) An optimum moisture content

Q117 — (b) Pore water carries the load and the soil shears/smears — *Wet soil shears/smears instead of densifying to maximum.*

Q136 — (b) Compaction of the subsurface to reduce percolation

Q137 — (a) A firmed surface can improve a following wheel's traction

Q138 — (b) Increase contact pressure and surface compaction

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10, pp. 253–258 (Table 10-3, first-pass compaction); Gill & Vanden Berg — USDA Agriculture Handbook 316; standard soil-physics texts (Proctor compaction, bulk density); ICAR-CIAE soil-compaction studies; NPTEL soil-dynamics lectures.